LSI Engine ECU(EControls) Software Request Form

ECU Spec. Sheet No.	SI21999	rev.	1

No.	Item	- Fill in below-
A	OEM Name	KEA
B	Machine Model	Standard Engine
C	Base Engine Model	WG1605

KAPA project number:	*****
Project start date:	2014/4/17
Application Engineer(or EGM)	person in charge
Revised date :	2021/5/17
Revised by :	Takahiro Tokunaga

Pass/Fail	Base Engine Spec.:Pass

File Extension	037

No.	Item	Details	Remarks	OEM Request	Rev.	Memo
1	Application		-	Standard Engine	1	
2	Fuel Type	Gasoline, LPG, NG	-	Gasoline	1	
3	Emission regulation#1	CARB/EPA	-	CARB/EPA	1	
4	Emission regulation#2	EU StageV	-	Not Required	1	
5	Catalyst	Type of Catalyst	-	Catalytic Muffler	1	
6	Side PTO		-	Not Required	1	
7	Max. Speed	3600 RPM	-	3600 rpm	1	
8	Min. Speed	800 RPM or Higher	-	800 rpm	1	
9	Load-Limit	Derate Engine power. ECU can have at least 3 power curves and switch those.	-	Not Required	1	
10	Max. Speed Limit Setpoints	Switch Engine Max. Speed Settings	-	Not Required	1	
11	Throttle Input Settings	Throttle Input Settings	Select or Fill in	TSC1	1	
		TSC1 Source Address	Select or Fill in	0xD0(Control)&0x0B(Derate)	1	
		Analog :				
		Control Mode	Speed or Throttle Control			
		Primary Potentiometer Characteristic	Single or Dual output			
		Sensor1 Output Spec. (Range:0.2~4.8V)	e.g.) 0.5-4.5V/0-100%			
		Sensor2 Output Spec. (Range:0.2~4.8V)	e.g.) 4.5-0.5V/0-100%			
		Secondary Potentiometer	Single output Only			
		Sensor Output Spec. (Range:0.2~4.8V)	e.g.) 0.5-4.5V/0-100%			
		Tap Up/Down :				
		Single-Tap Speed Step Change	e.g.) 100rpm			
		Button-Hold Speed Ramping Max Rate	e.g.) 500rpm/sec			
		Gov1/Gov2 Input Configuration	e.g.) +V=Active			
		Constant Speed SW (Max 4 Set Points): O:Active, −: Inactive				
		Set Point 1 (Gov1=−, Gov2=−)	Low Idle Speed	rpm		
		Set Point 2 (Gov1=O, Gov2=−)	e.g.) 1800	rpm		
		Set Point 3 (Gov1=−, Gov2=O)	e.g.) 2400	rpm		
		Set Point 4 (Gov1=O, Gov2=O)	e.g.) 2400	rpm		
		Gov1/Gov2 Input Configuration	e.g.) GND=Active			
12	Vehicle Speed Limit	Speed Limit Function	-	Not Required	1	
		Vehicle Speed Input to ECU	Describe detail From A			
		Speed Limit Setpoint	-	km/h		
		Speed Limit Enable SW	Hardwire SW			
CAN		Details	Remarks	OEM Request	Rev.	Memo
13	PGN List	Standard(see below) or Others	-	Standard	1	
14	EBC1(PGN61441)	Engine Auxiliary Shutdown (SPN970)	-	SA:Any	1	KEA uses 0x0B for EBC1
Inputs		Details	Remarks	OEM Request	Rev.	Memo
15	Interlock System	Starter Interlock	Inhibit Starter action	Not Required	1	
		Accelerator Interlock	Inhibit increasing speed	Not Required	1	
		Footpedal Idle Validation on Start	Inhibit increasing speed			
		Another Interlock	-	Not Required	1	

Outputs		Details	Remarks	OEM Request	Rev.	Memo
16	Output1	Select Item	e.g.) Oil Pressure Alert	Oil Pressure Alert	1	
		Select VIC PIN	e.g.) VIC1-E:AUX PWM2	VIC1-E: PWM2	1	
		Output Trigger	e.g.) Oil Pressure low	Oil Pressure low	1	
17	Output2	Select Item	e.g.) Coolant Temp. Alert	Coolant Temp. Alert	1	
		Select VIC PIN	e.g.) VIC2-C:AUX PWM5	VIC2-C: PWM5	1	
		Output Trigger	e.g.) >110℃(230°F)	>230°F	1	
18	Output3	Select Item	e.g.) Battery Voltage Alert	Battery Voltage Alert	1	
		Select VIC PIN	e.g.) VIC2-F:AUX PWM1	VIC2-F: PWM1	1	
		Output Trigger	e.g.) <10.0V	<10.0V	1	
19	Tachometer Output	Square-wave Output for Tachometer	Amplitude:0V-Vbat	Not Required	1	
		Tachometer Frequency (pulses/rev)	e.g.) 60pulses/rev			
20	Secondary Lockoff Valve *Current Limit: 4A(continuous), 5.8A(peak)	Additional Lock-off Valve	Required(VIC2-C) or Not Required			
DTC		Details	Remarks	OEM Request	Rev.	Memo
21	Flash Code MIL Blink	MIL Blink Function for Diagnosis	-	Not Required	1	
		Blink Config.(Blink on/off time, etc.)	-			
		Trigger Input	-			
22	Low Oil Pressure	Action	e.g.) Shutdown	No-action(MIL Only)	1	
23	Coolant Temperature High	Stage 1 Threshold	e.g.) >110℃(230°F)*Max118℃	>230°F	1	
		Stage 1 Latch/Unlatch time	e.g.) 20sec/30sec	20sec/30sec	1	
		Stage 1 Action	e.g.) Power Derate1	No-action(MIL Only)	1	
		Stage 2 Threshold	e.g.) >115℃/(240°F)*Max118℃	>250°F	1	
		Stage 2 Latch/Unlatch time	e.g.) 10sec/60sec	10sec/60sec	1	
		Stage 2 Action	e.g.) PD2 & Low Rev Limit	No-action(MIL Only)	1	
24	Intake Air Temperature High	Stage 1 Threshold	e.g.) >71℃ (160°F)	>160°F	1	
		Stage 1 Latch/Unlatch time	e.g.) 10sec/60sec	10sec/60sec	1	
		Stage 1 Action	e.g.) Power Derate1	No-action(MIL Only)	1	
		Stage 2 Threshold	e.g.) >88℃(190°F)	>190°F	1	
		Stage 2 Latch/Unlatch time	e.g.) 10sec/15sec	10sec/15sec	1	
		Stage 2 Action	e.g.) PD2 & Force I	No-action(MIL Only)	1	
25	Primary Potentiometer High/Low voltage	Sensor1 Short/Open Circuit Threshold	e.g.) H:4.8V、L:0.2V			
		Sensor2 Short/Open Circuit Threshold	e.g.) H:4.8V、L:0.2V			
		Latch/Unlatch time	e.g.) 0.5sec/3.0sec			
		Action	e.g.) PD1 & Low Rev Limit			
26	Secondary Potentiometer High/Low voltage	Short/Open Circuit Threshold	e.g.) H:4.8V、L:0.2V			
		Latch/Unlatch time	e.g.) 0.5sec/3.0sec			
		Action	e.g.) PD1 & Low Rev Limit			

WG1605	Gasoline		LPG		NG	
RPM	kW	Nm	kW	Nm	kW	Nm
800	7.7	91.9	8.1	97.4	7.5	90.0
1000	10.6	101.0	10.8	102.5	10.1	96.5
1200	13.3	105.6	13.4	106.4	12.5	99.6
1400	15.7	107.3	15.8	108.2	14.8	101.3
1600	18.4	109.7	18.2	108.6	17.1	102.3
1800	20.7	110.0	20.4	108.6	19.3	102.2
2000	23.2	110.7	22.8	109.1	21.3	102.1
2200	25.9	112.3	25.3	109.8	23.5	102.0
2400	27.9	110.9	27.2	108.2	25.4	101.0
2600	29.8	109.5	29.2	107.2	27.3	100.1
2800	31.7	108.0	31.1	106.0	29.0	98.9
3000	33.9	107.9	32.9	104.6	30.8	97.9
3200	35.8	106.8	34.6	103.2	32.3	96.3
3400	37.8	106.2	36.2	101.8	33.8	94.9
3600	39.5	104.8	37.6	99.8	35.3	93.6

Form A: Any detailed information, additional requests and questions

Broadcast (Tx PGN List) by 4G ECU: broadcast by the 4G ECU at regular time intervals
1) Oil Pressure Monitoring: PGN 65263, SPN 100 must be enabled → EFL/P1 (Engine Fluid Level/Pressure #1)
2) Engine Coolant/Air Temperature Monitoring: PGN 65262, SPN 110, 52 must be enabled → ET1 (Engine Temperature #1)
3) Engine Speed/Torque Monitoring: PGN 61444, SPN 513, 190 must be enabled → EEC1 (Electronic Engine Controller #1)
4) Fuel Rate Monitoring: PGN 65266, SPN 183 must be enabled → LFE1 (Fuel Economy (liquid))
5) Battery Voltage Monitoring: PGN 65271 must be enabled → VEP1 (Vehicle Electrical Power)
6) Engine % Load Monitoring: PGN 61443, SPN 92 must be enabled → EEC2 (Electronic Engine Controller #2)
7) Engine Hour Monitoring: PGN 65253, SPN 247 must be enabled → HOURS (Engine hours / revolutions)
8) Diagnostic Faults: DM1 (Active DTC's), DM2 (Historic DTC's)

Supported (Tx PGN List) by 4G ECU: Transmitted when the OEM control panel request these PGN's.
1) Include all broadcasted PGN's from the above selection.
2) ECU Part Number/ Serial Number for EOL Purpose: PGN 64965 must be enabled → ECU ID
3) Engine Make/Model Number/ Serial Number: PGN 65259 must be enabled → CI (Component ID - Make/Model Number)
4) ECU Software Identification: PGN 65242 must be enabled → SOFT (Software ID)

Required (Rx PGN List) by 4G ECU: CAN Message Required by OEM Control Panel at regular time intervals
1) Torque/Speed Control #1: PGN 11, SPN 695, 898, 518 must be enabled → TSC1
2) Engine Shut Down Request: PGN 61441, SPN 970 must be enabled → EBC1 (Electronic Brake Controller 1)
    a) Used for customer configurable shut downs.

Supported (Tx PGN List) by 4G ECU: Used to identify which Rx E-Control SPN will be supported by the 4G ECU.
1) Torque/Speed Control #1: must be enabled for TSC1 control purpose
    a) SPN 695 (Override Control Mode/Override Control Mode Priority)
    b) SPN 898 (Speed Limit)
    c) SPN 518 (Torque Limit)
2) Engine Shut Down Request: SPN 970; Source Address:11 must be enabled → EBC1 (Electronic Brake Controller 1)
    a) Used for customer configurable shut downs.

Standard PGN List

	Group Acronym	Decimal PGN	SPN	Parameter Name	Parameter
Broadcast	EEC2	61443	91	Accelerator Pedal Position 1	FPP1
			92	Engine Percent Load At Current Speed	torque_pct
			29	Accelerator Pedal Position 2	FPP2
	EEC1	61444	513	Actual Engine - Percent Torque	torque_pct_gro
			190	Engine Speed	speed
	EI	65170	1241	Mass Flow	J1939_mdot_f_ports
	EEC3	65247	514	Norminal Friction - Percent Torque	torque_pct_fric
			515	Engine's Desired Operating Speed	rpmd_gain
	EC1	65251	544	Reference Engine Torque	torque_Nm_ref
	HOURS	65253	247	Engine Total Hours of Operation	HM_RAM_secound
	ET1	65262	110	Engine Coolant Temperature	ECT
			52	Engine Intercooler Temperature	IAT
	EFL/P1	65263	100	Engine Oil Pressure	376 or 0kPa
	LFE1	65266	183	Engine Fuel Rate (Gasoline/LPG)	J1939_Vdot_f_ports
			51	Engine Throttle Valve 1 Position	TPS1
	IC1	65270	105	Engine Intake Manifold 1 Temperature	MAT
			106	Engine Air Intake Pressure	MAP
	VEP1	65271	168	Battery Potential/Power Input 1	Vbat
			158	Keyswitch Battery Potential	Vsw
On Request	ECUID	64965	2901	ECU Part Number	"E2452301D"
			2902	ECU Serial Number	"XXXXXX"
			2903	ECU Location	"Engine1" (fixed)
			2904	ECU Type	"ECM" (fixed)
	SOFT	65242	234	Software Identification	"EHXXX-699XX"
	CI	65259	586	Make *After EOL Test	"KBTA" (fixed)
			587	Engine Model *After EOL Test	"EGXXXXXXXX"
			588	Engine Serial Number *After EOL Test	"XXXXXXX"
			233	Unit Number	"1" (fixed)
	AMB	65269	108	Barometric Pressure	BP
KGST		65360	-	These PGNs are used for communication with	
EOL Tester		65361	-	KGST & EOL Tester	

Form B: PGN Request form

e.g.) Standard +

From C: Free Form

Software Revision (Software name)

No.	Date	Author	Subject	Description of Content	Setting state	Software rev.
1	2014.4.20	T.Tokunaga	1st Sample		Can't move to Mass	1
12	2021.5.17	T.Tokunaga	Final Check		Can move to Mass	12

Prototype software

	Approved by	Software name	Date
KEED	K.Shiraishi	4G_1.6L_KEA3_VER12_G.037	2021.5.17

Production-intent software

Software name : 4G_1.6L_KEA3_VER12_G.037.mot

Sales office Approved by	EGM Approved by	KEED2
Date:	Date:	Date:

ECU software preparation work flow

Prototype stage

Sales Office (or EGM)
Fill out this request form
Change the file name *e.g.)LSI_SRFv4_KBTA_WG1605_G_rev1_2021APR1
ECU Spec. Sheet No. : Blank
Sheet rev. : 1

→ Sheet →

KEED2
Make a sample software
Give a Spec. Sheet No. for management
ECU Spec Sheet No. : SI*****
Sheet rev. : 2

→ Sheet → mot →

Sales Office (or EGM)
Keep the request form
ECU Spec Sheet No. : SI*****
Sheet rev. : 2

If the ECU spec. is need to change

Updater(KEED2 or Sales Office/EGM)
Update the request form.
ECU Spec. Sheet No. : SI*****
Sheet rev. : must be updated

↔ Sheet mot ↔

Receiver
Check and keep the request form
ECU Spec. Sheet No. : SI*****
Sheet rev. : **

Production stage

Sales Office (or EGM)
Fill in Production-intent software name and sign.
Must not update Sheet rev.
ECU Spec. Sheet No. : SI*****
Sheet rev. : final no.

→ Sheet →

EGM
Check and sign
ECU Spec. Sheet No. : SI*****
Sheet rev. : final no.

→ Sheet →

KEED2
Check and sign
make mass-produciton software
ECU Spec. Sheet No. : SI*****
Sheet rev. : final no.

Sheet

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